

Approximating NSOP_1 Structures

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Smoothly Approximable Structures

Definition

A **smoothly approximable** structure is a countable ω -categorical structure $\mathcal{M}$ for which there is an ascending chain of finite homogeneous substructures $\mathcal{N}_i$ such that $M = \bigcup_{i < \omega} N_i$.

- Introduced by Lachlan, theory developed by Cherlin and Hrushovski
- Influential in the development of simple theories

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Main Idea

Build understanding of NSOP₁ as a generalization of simple by way of smooth approximability instead of forking independence.

Example

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- Over a finite field, this is smoothly approximable
- Approximated using a sequence of finite-dimensional subspaces
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Limitation

The definition of $\perp^\Gamma$ is specific to vector spaces and doesn't make sense outside of this context.

- Developed a nice abstract framework for approximation independent of a specific example, defining $\perp^{\text{lim}}$ as the limit of non-forking independence in a directed system of substructures
- Proved various properties of $\perp^{\text{lim}}$ that follow from simplicity of the approximating structures
- Showed T_∞ (mostly) fits into this framework, with $a \perp_C^{\text{lim}} B$ in T_∞ being equivalent to $a \perp_{\text{acl}(C)}^\Gamma \text{acl}(B) \text{acl}(C)$

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Limitation

Theoretically applied much more broadly, but still just the one example.

Generalized Approximation Framework (H-S)

Let $\mathcal{H}$ be a directed system of substructures of a structure $\mathcal{M}$. We say that a property holds “for large enough $\mathcal{N}$ ” to mean that there is some $\mathcal{N} \in \mathcal{H}$ such that for all $\mathcal{N}' \in \mathcal{H}$ with $\mathcal{N} \subset \mathcal{N}'$, the property holds in $\mathcal{N}'$.

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Definition

- A formula is *eventually true (false)* in $\mathcal{H}$ if it is true (false) for large enough $\mathcal{N}$ in $\mathcal{H}$.
- A partial type $\pi(x, b)$ *eventually forks (eventually doesn't fork)* over C if for large enough $\mathcal{N}$ in $\mathcal{H}$, $\pi(x, b)$ forks (doesn't fork) over $C \cap N$.
- A type $p(x)$ *eventually forks* over C if it contains a finite subtype that eventually forks over C . It *eventually doesn't fork* over C if every finite subtype eventually doesn't fork.
- $A \perp_C^{\lim} B$ if for each finite tuple $a \in A$ there is $\mathcal{N}$ large enough such that $a \perp_{\bar{C} \cap N}^{\mathcal{N}} \bar{B} \cap N$, where $\perp^{\mathcal{N}}$ is non-forking independence in $\mathcal{N}$.

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Given $\mathcal{M} \models T$ and $\mathcal{H}$ a directed system of substructures $\mathcal{N}$ of $\mathcal{M}$ we say $\mathcal{H}$ *approximates* $\mathcal{M}$ if conditions 1 through 5 hold. Conditions 6 and 7 are originally stated as optional.

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5. Strong finite character of $\not\vdash^{\lim}$: If $A \not\vdash_C^{\lim} B$, then there is $\varphi(x, b, c) \in \text{tp}^{\mathcal{M}}(A/\bar{B}\bar{C})$ which eventually forks over $\bar{C}$ in $\mathcal{H}$.

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7. Homogeneity: If $a \equiv_A^{\mathcal{M}} b$, then $a \equiv_{A \cap N}^{\mathcal{N}} b$ for all $\mathcal{N}$ containing a, b .

Examples

We show there are actually several examples of NSOP_1 theories approximated by simple structures in the sense of Harrison-Shermoen, suggesting that approximability is a relatively common property in NSOP_1 .

The following are all examples of NSOP_1 theories that can be approximated in this framework (more or less):

- T_∞
- T_{feq}^*
- Infinite-dimensional parameterized vector spaces
- ω -free PAC fields

Example

The generic theory of a parameterized equivalence relation. On an object sort O and a parameter sort P , the relation $E_x(y, z)$ is defined such that for each $p \in P$, $E_p(y, z)$ is an equivalence relation on O .

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- Approximating by restriction to finitely many parameters fails to satisfy convergence of truth value with the formula $(\forall x, y)(\exists p) \neg E_p(x, y)$.
- Approximate instead by substructures $\mathcal{M}_n$ where we restrict to at most n classes with respect to each parameter. This satisfies all conditions **except automorphism invariance**.

Automorphism Invariance vs Directed System

Automorphism invariance for T_{feq}^* is in tension with $\mathcal{H}$ being a directed system. In order to get automorphism invariance, we need to consider not just a single chain of substructures $\mathcal{M}_n$, but the collection of all substructures of $\mathcal{M}$ that are isomorphic to some $\mathcal{M}_n$.

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Solution

Take $\mathcal{H}$ to be the automorphism invariant collection of substructures and define $\downarrow^{\text{lim}}$ on a filtration of $\mathcal{H}$ to a directed system that satisfies the remaining axioms. As long as $\downarrow^{\text{lim}}$ is invariant of the choice of filtration, we can get all the same conclusions approximating via the filtration.

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- Forking is trivial in a structure where we restrict to finitely many classes, so limit independence along any filtration is trivial.

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- Forking is trivial in a structure where we restrict to finitely many classes, so limit independence along any filtration is trivial.
- This fixes the issue with automorphism invariance, so T_{feq}^* is approximable by the chain of $\mathcal{M}_n$'s.
- It also means $\downarrow^{\text{lim}}$ is trivial in the approximated theory. Since $\downarrow^{\text{lim}}$ inherits base monotonicity from forking in the approximation, even in cases where it has more structure it will not recover $\downarrow^K$ in strictly NSOP₁ theories.

Motivation: Dobrowolski's Dimension for T_∞

Definition

Given a definable set X over a model $M = \bigcup N_r$ (an approximating sequence in the sense of Granger) let $\dim(X)$ be the function $f : \omega \rightarrow \mathbb{Z}$ given by $f(r) = \text{rk}_{N_r}(X \cap N_r)$, where rk_{N_r} denotes Morley rank in N_r .

Dobrowolski showed that for any definable set X , $\dim(X)$ is almost everywhere equivalent to a linear function, which can then be ordered lexicographically as pairs (slope, intercept).

This notion of dimension does not recover $\perp^K$, but is the motivation behind our notion of limit dimension.

Limit Dimension

We would like to construct a notion of dimension constructed from the approximation that corresponds to $\perp^K$ in the limit. Dimension naturally gives rise to the following independence relation:

Definition

Let $\perp^{\dim}$ be the independence relation representing a drop in limit dimension:

$$A \perp_C^{\dim} B \iff \dim_C(A/C) = \dim_C(A/B)$$

In strictly NSOP₁ theories, $\perp^K$ does not satisfy base monotonicity. This means our dimension will be defined over a base set C .

Definition

Let $\mathcal{M} \models T$ be a structure approximated by $\mathcal{H}$. For $\mathcal{N} \in \mathcal{H}$ large enough, a base set $C \subseteq M$, and a set X definable over M , let $\dim_{C \cap N}^{\mathcal{N}}(X \cap N)$ be a notion of dimension valued in a partially ordered abelian group G .

We call $\dim_{C \cap N}^{\mathcal{N}}(X \cap N)$ the **approximated dimension** of X over C in $\mathcal{N}$.

The **limit dimension** of X over C is the equivalence class $[f]$ of functions $\mathcal{H} \rightarrow G$ almost everywhere equivalent to the function $f(\mathcal{N}) = \dim_{C \cap N}^{\mathcal{N}}(X \cap N)$ for large enough $\mathcal{N}$.

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In T_{feq}^* , the approximated dimension is a tuple $(\text{rk}_=(X), (\text{rk}_p(X), \text{deg}_p(X))_{p \in C})$ where $\text{rk}_=$ is Morley rank in $L = \{=\}$, rk_p is Morley rank in $\mathcal{N}_p$, and deg_p is the multiplicity.

Theorem

If for large enough $\mathcal{N}$ the approximated dimension satisfies invariance, algebraicity, union, fibration, additivity, and eventual uniform definability, then $\downarrow^{\dim}$ satisfies symmetry, monotonicity, strong finite character, and extension.

Limit Dimension Axioms

Theorem

If for large enough $\mathcal{N}$ the approximated dimension satisfies invariance, algebraicity, union, fibration, additivity, and eventual uniform definability, then $\downarrow^{\dim}$ satisfies symmetry, monotonicity, strong finite character, and extension.

Questions

- What additional axioms do we need to recover the independence theorem for $\downarrow^{\dim}$?
- Does the notion of parameterized dimension for T_{feq}^* also recover Kim-independence when applied to parameterized vector spaces?
- Is there a notion of dimension for T_∞ fitting this axiomatic framework?